

Software Requirements Specification

GridironIQ

A Multi-Agent Penalty Review and Rule Citation System for American Football

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Revision History

Version	Date	Author	Summary of Change
0.1	August 4, 2026	Joseph Hudson	Initial outline; scope and objectives drafted.
0.2	August 6, 2026	Joseph Hudson	Functional and non-functional requirements added.
1.0	August 8, 2026	Joseph Hudson	Diagrams, glossary, assumptions, and traceability added; baselined for submission.

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1. Project Title and Overview

1.1 Purpose of This Document

This Software Requirements Specification (SRS) defines the scope, functionality, quality attributes, and structural design of GridironIQ, the application I am building for my final project. It is written to be complete enough that the implementation phase can proceed directly from it, and specific enough that each requirement can be verified rather than argued about. Every requirement in Sections 3 and 4 is stated with a measurable condition or an acceptance test.

1.2 Problem Statement

Penalty enforcement is the most contested part of American football officiating. Officials make judgment calls in roughly one second, from a single vantage point, on plays involving twenty-two moving players. Replay review helps, but it introduces its own problems: reviews are slow, the standard for overturning a call is inconsistently applied, and the reasoning behind a decision is rarely captured in a form anyone can audit afterward. The result is a recurring credibility problem for the sport and no usable record for training or evaluating officials.

GridironIQ addresses the narrower and more tractable part of that problem. It does not attempt to replace an official's judgment. It provides a second set of eyes that works consistently, explains itself, and cites the rule it is applying. For a given play, the system extracts evidence frames, routes them to a set of specialized penalty agents, aggregates their findings into a single confidence-scored verdict, attaches the governing rule text, and presents all of it to a human official who makes the final call. Every step is logged.

1.3 Target Audience

- Replay Officials and on-field officiating crews, who use the system during review to see a second opinion with its supporting evidence and rule citation.
- Officiating supervisors and evaluators, who use the accumulated decision record to assess crew consistency and identify rules that are being applied unevenly.
- Broadcast and analytics staff, who need a fast, explainable read on a contested call during a game.
- Coaching and player-development staff at levels below the professional game, where full replay infrastructure is not available but recorded film is.

1.4 Product Perspective and Rationale

GridironIQ is a self-contained application that depends on three external services: a source of play video and play-by-play metadata, a large language model inference endpoint, and an indexed copy of the rulebook. It grew out of FlagAI, an earlier multi-agent prototype of mine that produced penalty calls but had no way to justify them. That gap is the reason this system exists: an unexplained machine verdict is worthless in officiating, because an official cannot act on a conclusion they cannot defend. The architectural consequence is Section 4's explainability requirement (NFR-5), which forbids the system from displaying any verdict that is not accompanied by its evidence and its rule citation.

2. Scope and Objectives

2.1 System Boundaries

The system boundary encloses ingestion, frame extraction, agent orchestration, rule retrieval, adjudication, reporting, and audit logging. Video capture, the language model itself, and the authoritative rulebook source are outside the boundary and are treated as external interfaces, shown as external systems in the use case diagram in Section 5.

2.2 In Scope

- Ingestion of recorded play clips with structured game-situation metadata.
- Automated segmentation of a clip into play phases and extraction of labeled key frames.
- Analysis by at least three specialized penalty agents, each returning a structured, confidence-scored finding.
- Retrieval of the governing rule citation for every flagged finding.
- A review interface allowing an authenticated official to confirm, override, or defer each flag.
- Categorized capture of override reasons for later evaluation of agent accuracy.
- An append-only audit log covering every ingestion, agent invocation, flag, and decision.
- Per-game report generation and export to PDF or CSV.
- Role-based access control across four user roles and configuration of the active rule-set version.

2.3 Out of Scope

- Live in-game officiating. The system operates on recorded clips only and issues no ruling of its own; it is decision support, not a decision maker.
- Automatic enforcement of any call. No output of the system changes a game state without a human official's confirmation.
- Player tracking hardware, chip telemetry, or any sensor integration. The system works from video and metadata alone.
- Real-time broadcast overlay or graphics generation.
- Integration with official league systems or use of proprietary league data. The prototype uses publicly available and self-recorded footage.
- Fouls outside the three implemented categories in version 1.0. The agent interface is designed for extension, but additional categories are future work.
- Native mobile applications. The interface is browser-based and responsive.

2.4 Objectives

ID	Objective	Measure of Success
OBJ-1	Produce a consistent, repeatable penalty assessment for a recorded play.	The same clip analyzed five times yields the same verdict at least four times, with confidence variance under 0.10.

ID	Objective	Measure of Success
OBJ-2	Make every recommendation explainable and defensible.	100% of flags presented to an official include evidence frames, agent rationale, and a rule citation.
OBJ-3	Keep review fast enough to be useful.	Verdict available within 45 seconds of ingestion at the 95th percentile (NFR-1).
OBJ-4	Preserve human authority over every call.	No game-affecting output is emitted without an official's confirmation; overrides are always available and always recorded.
OBJ-5	Create an auditable record suitable for officiating evaluation.	Every decision is reconstructable from the audit log, including the rule-set version in force at the time.
OBJ-6	Allow the foul catalog to grow without redesign.	A new penalty agent can be added by implementing one interface and registering it, with no change to the orchestrator (NFR-6).

3. Functional Requirements

The following requirements define what the system shall do. Each is uniquely identified, independently verifiable, and traced to at least one project objective in Section 9. Priority reflects implementation order: High requirements constitute the minimum viable system for the final project demonstration.

ID	Requirement	Specification and Acceptance Criteria	Priority
FR-1	Play clip ingestion	The system shall accept a play clip (MP4/MOV, ≤ 60 seconds, ≤ 500 MB) together with a metadata record containing game ID, quarter, game clock, down, and distance. Invalid format, oversize, or missing-metadata submissions shall be rejected with a specific error message within 3 seconds.	High
FR-2	Play segmentation and key-frame extraction	The system shall segment each ingested clip into pre-snap, snap, and post-snap phases and extract no fewer than 8 and no more than 24 key frames per play, storing each frame with its millisecond offset and phase label.	High
FR-3	Multi-agent penalty analysis	The system shall dispatch each play to at least three specialized penalty agents (offensive holding, pass interference, and pre-snap fouls). Each agent shall return a structured Finding containing foul type,	High

ID	Requirement	Specification and Acceptance Criteria	Priority
		offending side, confidence score (0.00–1.00), and a written rationale.	
FR-4	Rule citation retrieval	For every Finding at or above the flag threshold, the system shall retrieve and attach at least one governing rule citation (rule, section, article, and excerpt) from the indexed rulebook. A Finding without a citation shall not be presented as a flag.	High
FR-5	Confidence scoring and flag thresholding	The system shall aggregate agent Findings into a single play verdict and raise a flag only when the aggregated confidence meets or exceeds a configurable threshold (default 0.70). Findings below the threshold shall be retained as advisory and marked as such.	High
FR-6	Review and adjudication interface	The system shall present each flagged play to an authenticated Replay Official with the evidence frames, rationale, and rule citation, and shall allow the official to confirm, override, or defer the call in no more than three interactions.	High
FR-7	Official feedback capture	When an official overrides a flag, the system shall require a categorized reason (missed evidence, misapplied rule, poor video angle, or other) and shall store the override with the original Finding for later model evaluation.	Medium
FR-8	Immutable decision audit log	The system shall write an append-only audit entry for every ingestion, agent invocation, flag, and official decision, recording actor, action, payload hash, and UTC timestamp. Audit entries shall not be editable or deletable through the application interface.	High
FR-9	Game report generation and export	The system shall generate a per-game report summarizing every reviewed play, verdict, confidence score, rule citation, and official decision, and shall export it as PDF or CSV in under 30 seconds for a game of up to 180 plays.	Medium
FR-10	Authentication and role-based access	The system shall authenticate all users and enforce four roles — Replay Official, Officiating Supervisor, Broadcast Analyst, and System Administrator — such that only Supervisors and Administrators may change the flag threshold or active rule-set version.	High

ID	Requirement	Specification and Acceptance Criteria	Priority
FR-11	Season rule-set configuration	The system shall allow an authorized user to select the active rule-set version for a game, and shall stamp every Finding and citation with the rule-set version used, so that prior analyses remain reproducible after a rule change.	Medium

4. Non-Functional Requirements

These quality attributes constrain how the system behaves rather than what it does. Each states a measurable target and the method by which it will be verified.

ID	Attribute	Requirement	Verification Method
NFR-1	Performance	The system shall return a complete multi-agent verdict for a 15-second play clip in 45 seconds or less at the 95th percentile, measured from ingestion acknowledgement to verdict availability, for up to 5 concurrent analyses.	Load test of 100 sequential plays and 5 parallel sessions; timing recorded from application logs.
NFR-2	Reliability and availability	The analysis service shall maintain 99.0% availability during a scheduled review session. If an individual agent fails or times out, the orchestrator shall return the remaining Findings with a degraded-analysis notice rather than failing the play.	Fault-injection test in which one agent is forcibly failed; system must still produce a verdict.
NFR-3	Usability	A trained official shall complete the review of a flagged play in 90 seconds or less without written instructions. The review interface shall meet WCAG 2.1 Level AA for contrast, keyboard navigation, and focus order.	Timed task walkthrough with 3 test users; automated accessibility scan with zero Level AA violations.
NFR-4	Security	All traffic shall use TLS 1.2 or higher, stored clips and audit records shall be encrypted at rest, credentials shall be hashed with a memory-hard algorithm, and no API key shall appear in source control or client-side code.	Secret-scanning on every commit; manual review of transport and storage configuration.
NFR-5	Explainability and traceability	Every flag presented to an official shall be accompanied by the contributing agent, its confidence score, the evidence frames used, and the governing rule citation. No verdict shall be displayed without this supporting record.	Inspection of 20 randomly sampled flags; all four elements must be present.
NFR-6	Maintainability and portability	The system shall run from a single container definition on Windows, macOS, and Linux with no code changes, and a new penalty agent shall be addable by implementing one interface and registering it, without modifying the orchestrator.	Clean-machine deployment on all three platforms; addition of a demonstration agent with no orchestrator edits.

5. Use Case Model

5.1 Actors

Actor	Description and Responsibility
Replay Official (primary)	Submits or selects plays for review, examines flagged calls with their supporting evidence, and confirms, overrides, or defers each recommendation. Holds final authority over every call.
Officiating Supervisor	Reviews decisions across a game or crew, generates game reports, and configures the active rule-set version and flag threshold.
Broadcast Analyst	Submits plays for analysis and retrieves generated reports; has read access to verdicts but cannot adjudicate.
System Administrator	Manages user accounts, role assignments, and system configuration. Has no adjudication authority.
Video and Play-by-Play Feed (external)	Supplies the play clip and the structured game-situation metadata consumed at ingestion.
LLM Inference Service (external)	Executes the vision-and-language reasoning invoked by each penalty agent.
Rulebook Knowledge Base (external)	Provides the indexed rule text from which citations are retrieved.

5.2 Use Case Diagram

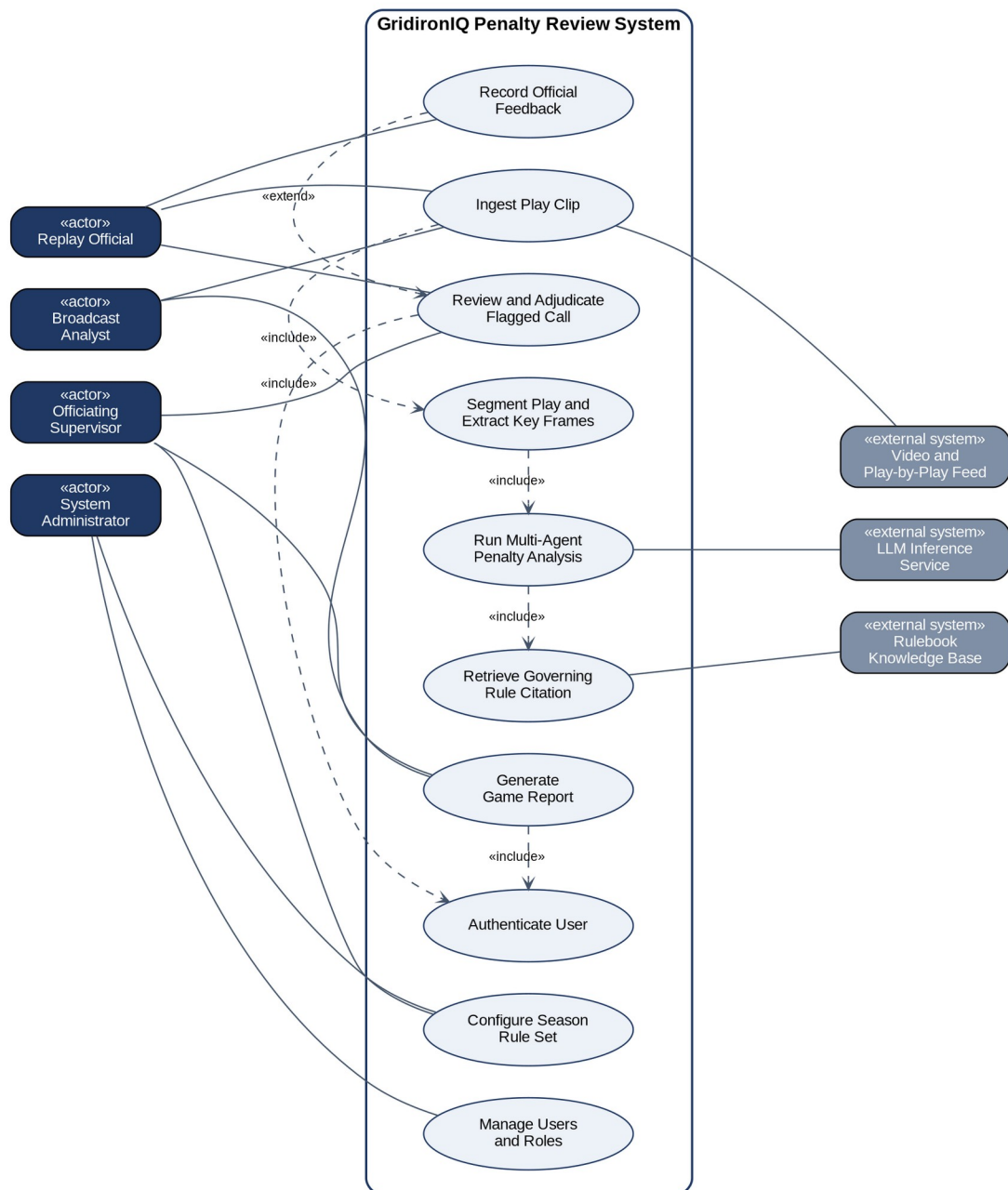


Figure 1. GridironIQ use case diagram, showing primary actors, external systems, and the system boundary.

5.3 Expanded Use Case: Review and Adjudicate Flagged Call

Field	Detail
Use case ID	UC-05 (satisfies FR-6, FR-7, FR-8)
Primary actor	Replay Official
Preconditions	The official is authenticated; the play has completed analysis and produced at least one Finding at or above the flag threshold.
Trigger	The official opens a play marked as flagged in the review queue.
Main flow	1. System displays the play clip, the evidence key frames, and the aggregated verdict. 2. System displays the contributing agent, its confidence score, and its written rationale. 3. System displays the governing rule citation with its excerpt. 4. Official confirms or overrides the call. 5. System records the decision, writes an audit entry, and returns the official to the queue.
Alternate flow A	Official selects Defer. System returns the play to the queue marked as unresolved and writes an audit entry; no verdict is recorded.
Alternate flow B	Official selects Override. System requires a categorized reason before the decision is accepted (FR-7), then proceeds from step 5.
Exception flow	Analysis was degraded because an agent failed or timed out (NFR-2). System displays a degraded-analysis notice with the partial findings; the official may adjudicate or defer.
Postconditions	A decision record exists and is linked to the originating Finding; a corresponding audit entry has been written and cannot be modified through the application.

6. UML Class Diagram

The structural model is presented in two views. Figure 2 shows the domain model — the entities the system stores and the relationships among them. Figure 3 shows the agent subsystem, where the analysis behavior lives. Splitting the model this way keeps each view legible and reflects the actual separation in the codebase between persistence and reasoning.

The diagrams use standard UML notation: a filled diamond marks composition, where the part cannot exist without the whole; an open diamond marks aggregation; a hollow triangle marks generalization from an abstract parent; and a dashed arrow marks a dependency, labeled with the stereotype describing it. Multiplicities are shown at each end of an association.

Three design decisions visible in these models are worth stating explicitly.

- PenaltyAgent is abstract, and each foul category is a concrete subclass. This is what makes OBJ-6 achievable: adding a foul category means adding a subclass and registering it with the orchestrator, not editing the orchestrator itself.
- Finding aggregates one or more RuleCitation objects, so the explainability requirement (NFR-5) is enforced by the data model rather than by convention in the interface layer. A Finding cannot be persisted as a flag without its citation.
- OfficialDecision is a separate class from Finding rather than a status field on it. The machine's conclusion and the human's ruling are recorded independently, which is what allows override rates to be measured per agent (FR-7) and what guarantees the audit record shows both.

6.1 Domain Model

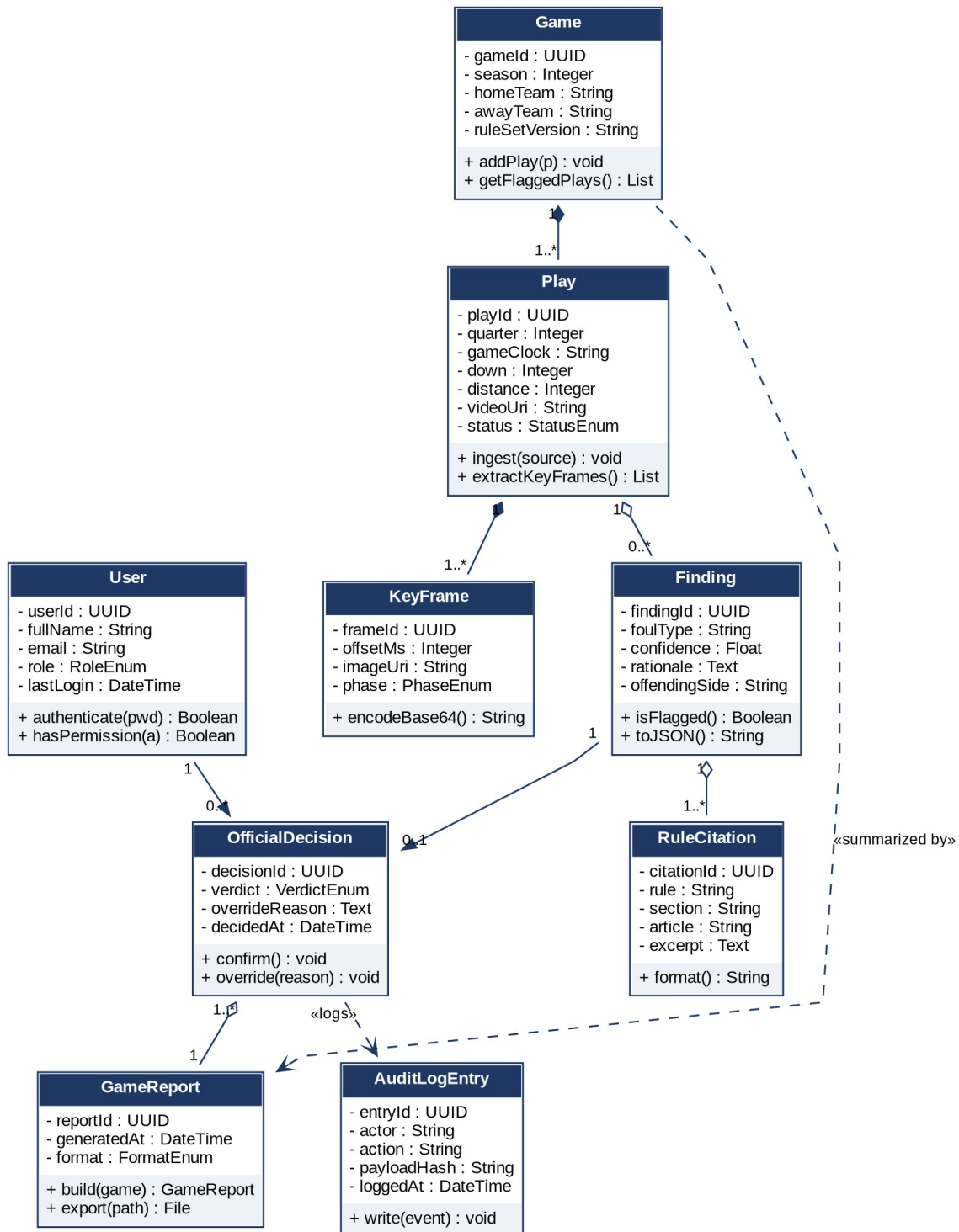


Figure 2. GridironIQ domain model, showing persistent entities, multiplicities, and the decision and audit trail.

6.2 Agent Subsystem

The orchestrator holds a registry of PenaltyAgent instances and knows nothing about any specific foul. Each concrete agent implements a single operation, `analyze()`, and returns a `Finding`. RulebookIndex is invoked separately to attach citations, so retrieval can be re-run or re-indexed without re-running analysis.

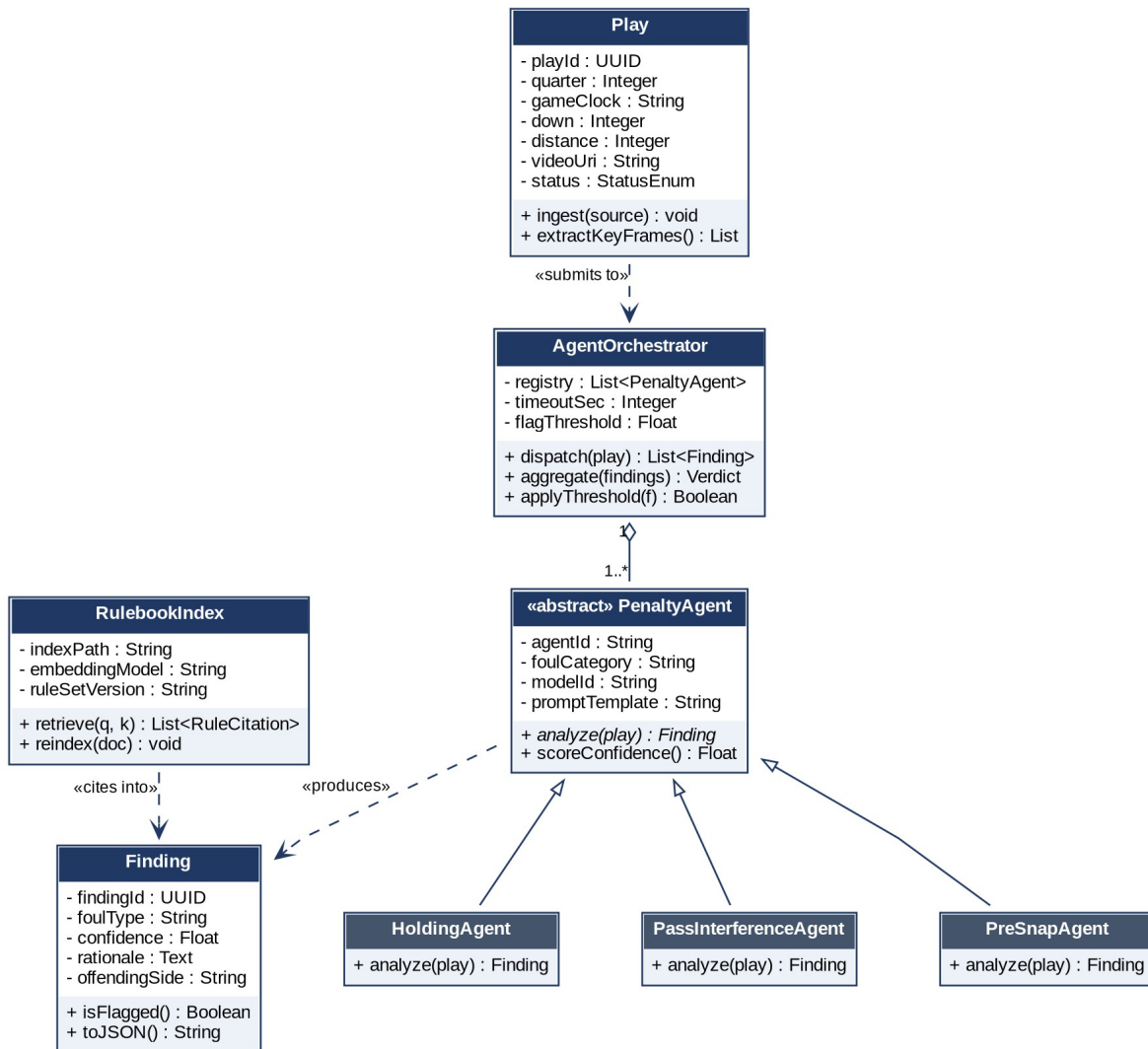


Figure 3. GridironIQ agent subsystem, showing the abstract agent interface and its concrete implementations.

7. Glossary

Term	Definition
Agent	A specialized analysis component responsible for one category of foul. Each agent has its own prompt template, evidence requirements, and confidence scale.
Confidence score	A value from 0.00 to 1.00 produced by an agent expressing how strongly the visual and contextual evidence supports the foul it has identified.
Finding	A structured result from a single agent: foul type, offending side, confidence score, rationale, and the evidence frames the agent relied on.
Flag	An aggregated Finding whose confidence meets or exceeds the configured threshold and which is therefore presented to an official for adjudication.
Flag threshold	The configurable confidence value, default 0.70, at or above which a Finding is escalated to a flag.
Key frame	A still image extracted from a play clip at a specific millisecond offset and labeled with its play phase (pre-snap, snap, or post-snap).
Orchestrator	The coordinating component that dispatches a play to the registered agents, applies the flag threshold, and aggregates Findings into a single verdict.
Play clip	A short video segment covering one football play, from the pre-snap formation through the end of the down.
Retrieval-augmented generation (RAG)	A technique in which relevant source text — here, rulebook passages — is retrieved from an index and supplied to a language model so its output can cite authoritative material.
Rule-set version	The identifier of the rulebook edition in force for a given game, recorded with every Finding so past analyses remain reproducible after rule changes.
Replay Official	The primary user of the system: the person who reviews flagged plays and confirms, overrides, or defers the recommended call.
Verdict	The system's aggregated conclusion for a play: no foul, advisory, or flagged, together with the supporting Findings.

8. Assumptions, Dependencies, and Constraints

8.1 Assumptions

- Play clips are of sufficient resolution and framing that the relevant players are visible; the system is not required to compensate for obstructed or missing camera angles.

- Users of the review interface understand football rules and terminology. The system assists officiating judgment; it does not teach the rulebook.
- The rulebook text used for citation retrieval is current for the configured rule-set version and is supplied in a machine-readable form.
- Reviews are conducted on recorded clips in a session, not against a live game clock.
- A single official adjudicates a given play; concurrent adjudication of the same play by multiple officials is not a supported workflow in version 1.0.

8.2 Dependencies

- Availability and rate limits of the external LLM inference endpoint, which govern the concurrency ceiling stated in NFR-1.
- Availability of the video source and the accuracy of the accompanying play-by-play metadata.
- A vector index over the rulebook corpus for citation retrieval.

8.3 Constraints

- Development is a solo effort within a single academic term, which is the reason version 1.0 implements three penalty categories rather than the full foul catalog.
- No proprietary league footage, telemetry, or officiating data is used. All test material is publicly available or self-recorded, and the rulebook is referenced for academic purposes only.
- The system runs on commodity hardware without a dedicated GPU; all model inference is performed through an external API.
- The system is a decision-support prototype and is explicitly not represented as fit for use in a sanctioned game.

9. Requirements Traceability

The matrix below confirms that every objective is served by at least one requirement and that no requirement exists without a purpose.

Objective	Satisfied By	Verified By
OBJ-1	FR-2, FR-3, FR-5, FR-11	Repeat-analysis consistency test
OBJ-2	FR-3, FR-4, FR-6, NFR-5	Sampled inspection of 20 flags
OBJ-3	FR-2, FR-5, NFR-1	Load and timing test
OBJ-4	FR-6, FR-7, FR-10	Workflow walkthrough of all output paths
OBJ-5	FR-8, FR-9, FR-11	Full game record rebuilt from audit log
OBJ-6	FR-3, NFR-6	Demo agent added; orchestrator unchanged